

**Minor Project Report
ANALOG CIRCUITS (UES301)**

Group Project Report File

**PROJECT TITLE: ASCIS
(Arduino based sensor controlled irrigation
system for small and medium sized farms)**

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(July-December 2024)**

DECLARATION

We, Rachit Saini, Debojeet Dutta, Yatin Jha and Meitreya Priyadarshi, declare that this project report is our own original work, carried out under the supervision of Dr. Sangeeta Kamboj in the ADC Laboratory. The statements made and conclusions drawn are an outcome of our own project work. We have not copied or plagiarized any part of this report from any other source. We have followed the guidelines provided by the Institute in writing this report. We have not submitted this project to any other university for the award of any degree.

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INTRODUCTION

India's economy is deeply rooted in agriculture, with a significant portion of the population dependent on farming for livelihood. However, fluctuating climatic conditions, soil degradation, and inefficient water management present ongoing challenges to sustainable farming practices. In recent years, India has witnessed remarkable growth in technology, particularly in the field of electronics and automation, creating opportunities to revolutionize traditional farming methods. This ignites the need to attempt the integration of an Arduino-based sensor-controlled irrigation system as a practical solution to address these challenges.

The system utilizes a combination of soil moisture sensors, temperature sensors, and humidity sensors etc. to monitor environmental conditions in real-time. Based on sensor readings, the Arduino microcontroller automatically controls the irrigation process by activating or deactivating water pumps, ensuring that crops receive the appropriate amount of water. The system is designed to be cost-effective, energy-efficient, and easy to implement. Additionally, it allows for manual override via a user-friendly interface, providing flexibility for farmers. Field testing demonstrates the system's effectiveness in conserving water while maintaining optimal soil moisture levels, thus promoting sustainable farming practices. The proposed system offers a scalable solution for automating irrigation, improving farm efficiency, and addressing the challenges posed by climate variability.

By automating water distribution based on precise environmental data, the system adapts to India's diverse and rapidly changing climatic conditions – where seasons and weather patterns can shift multiple times a year. This approach ensures optimal water use, increases crop productivity, and minimizes resource wastage, all while using the same soil and land. This attempt showcases how such electronic controllers can transform agricultural practices, fostering increased yield from existing farmland without the need for extensive manual intervention. This shift highlights India's potential to bridge the gap between traditional farming and modern technology, promoting sustainable agriculture and food security in the face of climate variability.

This report thoroughly explores the capabilities of Arduino Uno R3, focusing on foundational experiments to familiarize users with the inputs, outputs, coding, and interfacing components like sensors and resistors. Through systematic experimentation, students learn Arduino programming and circuit design, enhancing their understanding. Various topics, including analog/digital inputs/outputs and sensor data acquisition, are covered, preparing participants for complex tasks.

Additionally, the report outlines applying learned knowledge to practical scenarios, such as constructing an irrigation system to automate watering plants in regular farming techniques.

OBJECTIVES

The implementation of ASCIS would potentially help the farmers of the country to monitor their crop growth and take appropriate responsive measures timely to avoid crop failure. ASCIS helps to ace the production of multiple crops around different seasons of the year from the same farm land, by taking care of some vital physical and chemical requirements in the farm. As it is proven that more crop production from the same farm will enormously bring fortune to the farmers without actually using any hazardous chemicals, just by monitoring some requirements with the help of technology. This system has the potential to bring a major change in the agriculture sector with minimal technological knowledge and operational cost by appropriate support of the governments.

The main objectives of the invention are to set and adjust watering schedules based on real-time weather data in order to prevent over- or under-watering. Soil moisture sensors are used to water only when the soil reaches a specific dryness level, which is ideal for areas where water conservation is a priority. Multiple types of sensors can be integrated, often together with machine learning algorithms, to optimize water usage. The system can be controlled remotely and can provide detailed analytics on water usage and plant health. Finally, it can save the environment while the farmers are benefited at the same time.

We can also integrate this system with a drip irrigation system as well as a center pivot irrigation system. This system will help those farmers who have large farms, as it will decrease the labor cost, and we can even integrate automatic fertilizing systems.

CIRCUIT DIAGRAM

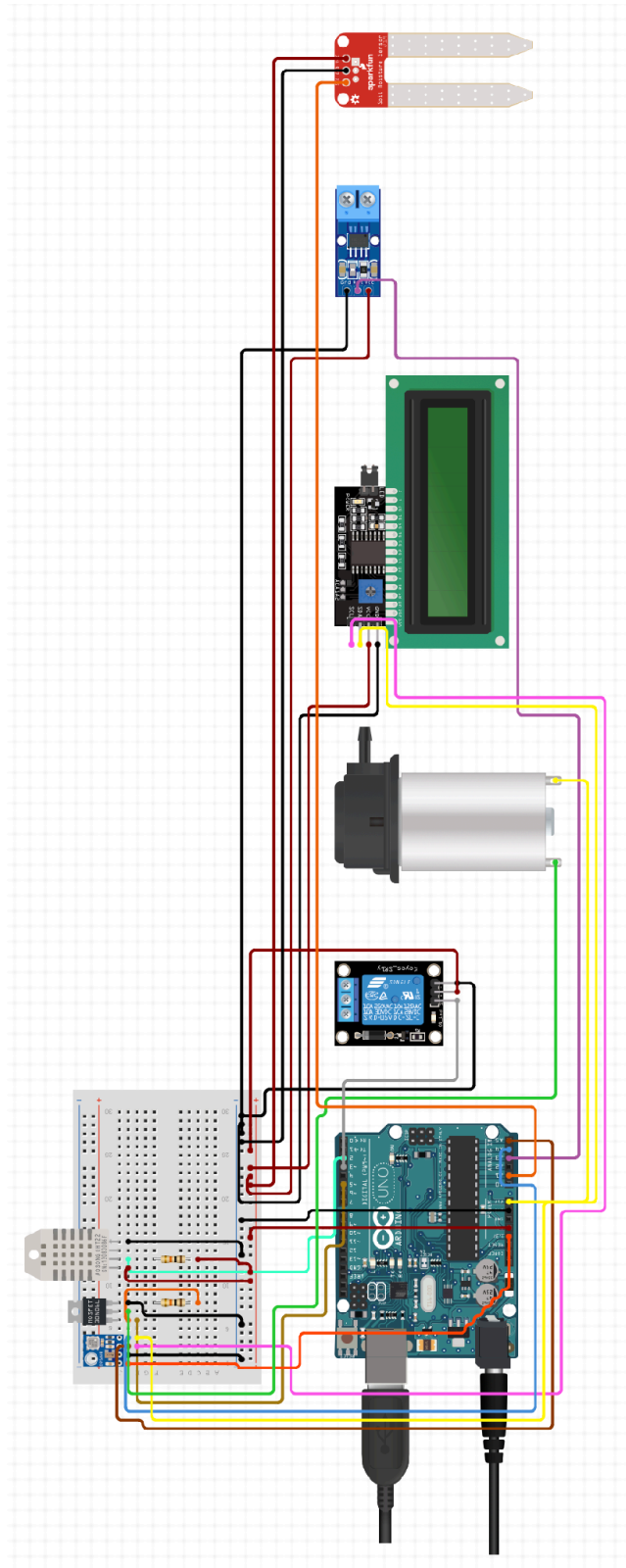


Figure 1. Circuit diagram of the ASCIS

WORKING PRINCIPLE

Key Components

1. Technology Infrastructure Set-up:

- Owners have to set up several sensors in the farm and place microcontrollers, valves and pumps that are controlled by them, integrated with the existing pipe networks and sprinklers.

2. Water Pipe Networks:

- A network of water pipes reaching all parts of the farm and covering almost all possible vicinities to ensure uniformity.

3. Real-Time Updates:

- Owners receive data about the local temperature, climate, soil conditions, etc.

Technology Used in this System

The ASCIS leverages a variety of technologies to ensure adequate water is supplied to the crops, along with other measures. An overview of the key technologies involved is given below.

1. Controllers:

- **Central Units:** These units manage the time and quantity of water distributed. They can be programmed based on time, soil moisture, weather conditions, or specific plant needs.

2. Sensors:

- **Capacitive Soil Moisture Sensor:** Measures the moisture level in the soil and signals the system to water only when necessary. Multiple units of these are required in the farm to adjust discrete water levels dissolved in the soil.
- **Humidity and Temperature Sensor DHT22/11:** Records the moisture content present in the air and the temperature of the local area of the farm, which helps the system decide the amount of water to be delivered. May require multiple installations to ensure accuracy in data.
- **Barometric Pressure Sensor BMP180:** A key component for monitoring climatic conditions; it serves as a module to predict the weather with the help of internet access given to the system.

3. Valves and Sprinklers:

- **Valves:** Control the flow of water to different parts of the irrigation system.

4. Pumps:

Used in cases where water needs to be drawn from a well, pond, or other sources. The pump is controlled by the system to deliver the water with the required pressure.

5. Arduino Microcontroller:

- **ATmega328P-based Arduino Uno:** The Arduino Uno with the ATmega328P provides an affordable, scalable, and user-friendly platform for building an efficient, sensor-controlled irrigation system.

6. Power Supply:

- **DC Power Supply:** Any 12 V wall DC power supply can be used to deliver power so that the system works properly. Using solar power installed on the farms would be even better.

7. **LCD Screen:** Displays the data received from the different sensors. It shows the numerical values of the conditions around the system in the atmosphere.

8. **Step-Down DC-DC Buck Converter LM2596:** The buck converter uses a switching transistor, inductor, diode, and capacitor to convert the input voltage to a lower output voltage. The switch rapidly turns on and off, creating a pulsating current that is then smoothed out by the inductor and capacitor to produce a steady, lower DC output. It converts the input voltage from 12 V to 5 V efficiently, minimizing power loss.

9. **Relay Module:** A relay module is an electronic component that uses an electromagnet to switch electrical circuits on or off. Relay modules allow low-voltage control systems (e.g., Arduino, Raspberry Pi, PLCs) to control high-voltage devices like AC appliances, motors, lights, and heaters. Relays provide galvanic isolation between the control circuit (low voltage) and the load circuit (high voltage). This isolation protects sensitive control electronics from high-voltage spikes and noise, helping to protect microcontrollers or other control units from potential damage.

10. **Voltage Regulator L7805CV:** The L7805CV is a linear voltage regulator that provides a fixed +5 V DC output from a higher input voltage. It is commonly used in electronics projects to supply a stable 5 V to components such as microcontrollers, sensors, and other digital devices.

Apart from these, several capacitors of 100 μ F, resistors of varying resistance, and jumper wires for connections are used as needed to make the system function as intended.

RESULTS

1. Increased Access to Irrigation Resources

- Users will have a wider range of irrigation sources available for drawing water, as reduced water wastage will save cost for the farm owners to migrate to a better quality water service.

2. Enhanced Farming Experience

- An automated interface and efficient irrigation system will lead to higher user satisfaction rates, motivating the introduction of more sensor applications and conditional predictions using algorithms based on the data collected.

3. Economic Growth for Farm Owners

- Increased crop produce or less crop failure will lead to increased sale of crops, and at the same time profits are scaled with the help of lower wastage costs.

4. Improved Farming Quality

- Farmers can expect better produce from the farm while at the same time conserving water and reducing costs.

5. Community Engagement

- The system will encourage more farm owners to use this technology and will develop a network among them.

6. Data-Driven Insights

- Analytics will provide valuable insights into user behavior, enabling strategic decision-making for future technology incubation.

APPLICATIONS

1. Smart Farming for Small and Mid-Sized Farms:

- The system is designed to optimize water usage and enhance crop yield through automated irrigation. It can be particularly useful for small and mid-sized farms, where water conservation and efficient resource management are critical.

2. Greenhouses and Controlled Environment Agriculture:

- In greenhouses, maintaining optimal moisture, temperature, and humidity is crucial. The ASCIS can automatically monitor these conditions using sensors and adjust the irrigation levels, ensuring ideal growth conditions for plants.

3. Home Gardens and Urban Farming:

- For home gardens and urban farming setups, this system can automate watering, reduce manual labor, and ensure plants receive the correct amount of water. It is especially beneficial for people who might not have time for daily plant care.

4. Precision Agriculture:

- The project integrates multiple sensors like soil moisture, temperature, humidity, and pH level sensors. This makes it suitable for precision agriculture, where farmers can monitor specific parameters and make data-driven decisions to enhance productivity.

5. Water-Sensitive Areas and Drought-Prone Regions:

- In areas with limited water resources or frequent droughts, ASCIS can significantly reduce water usage by ensuring that irrigation only occurs when necessary, thus preventing overwatering and conserving water.

6. Integration with Drip and Center Pivot Irrigation Systems:

- The system can be integrated with advanced irrigation methods such as drip irrigation and center pivot irrigation systems, which are efficient in water usage and are commonly used in large farms. This integration can further enhance the precision and effectiveness of water delivery.

7. Hydroponics and Vertical Farming:

- In hydroponic systems, where plants are grown without soil, maintaining the right water and nutrient levels is essential. ASCIS can be adapted to monitor and regulate water flow, pH levels, and other parameters in such setups.

8. Horticulture and Nursery Management:

- In horticulture and plant nurseries, where a variety of plants require different watering schedules, the system can help automate irrigation based on the specific needs of each plant type, reducing labor costs and improving plant health.

9. Remote Monitoring and Management of Farmlands:

- The system's integration with wireless communication modules (like Bluetooth and XBee) allows farmers to remotely monitor and control irrigation using smartphones or computers, making it useful for managing large or distant farmlands.

10. Community Gardens and Public Parks:

- Public parks and community gardens can benefit from automated irrigation systems to maintain greenery without the need for constant manual intervention. This helps in reducing maintenance costs and ensures efficient use of water resources.

11. Research and Educational Purposes:

- The project can be used as a practical educational tool for students and researchers in agricultural and environmental sciences, demonstrating the application of electronics and automation in sustainable farming practices.

12. Climate-Adaptive Agriculture:

- By using sensors like temperature and barometric pressure sensors, the system can adapt to varying climatic conditions. It is suitable for regions where weather patterns are unpredictable, helping farmers adjust irrigation based on real-time data.

13. Agricultural IoT (Internet of Things) Systems:

- With the potential integration of IoT devices, the system can collect data and use machine learning algorithms to predict the best irrigation schedules, optimize water use, and enhance overall farm management.

14. Smart City Initiatives:

- As part of smart city projects, this automated irrigation system can be implemented in urban green spaces, public parks, and green belts, contributing to sustainable urban development.

15. Organic Farming:

- The system can be particularly useful in organic farming, where precise water management is required without the use of chemical fertilizers. It helps maintain soil quality and supports the principles of sustainable agriculture.

These diverse applications highlight the potential of ASCIS to transform traditional irrigation methods, making agriculture more efficient, sustainable, and adaptive to modern challenges.

REFERENCES

1. Banzi, M. (2009). *Getting Started with Arduino*. O'Reilly Media, Inc.
2. Blum, J. (2013). *Exploring Arduino: Tools and Techniques for Engineering Wizardry*. John Wiley & Sons.
3. Michael, A. M., and Ojha, T. P. (2012). *Principles of Agricultural Engineering Volume II*. Jain Brothers.
4. Stout, B. A. (1982). *Energy Use and Management in Agriculture*. Springer-Verlag.
5. López, R., et al. (2019). "Irrigation control systems: A review." *Agricultural Water Management*, 217, 212–227.
6. Wolfert, S., Ge, L., Verdouw, C., & Bogaardt, M. J. (2017). "Big data in smart farming – a review." *Agricultural Systems*, 153, 69–80.
7. Priya, C., & Soundarya, P. (2017). "Arduino Based Smart Drip Irrigation System Using IoT." *International Journal of Advanced Research in Computer and Communication Engineering*, 6(3), 543–547.
8. Sharma, T., & Singh, A. (2020). "Arduino based automated irrigation system." *International Journal of Recent Technology and Engineering (IJRTE)*, 8(5), 546–551.
9. Jha, S. N. (2011). *Sensors in Agriculture and Food Quality Monitoring*. Springer.
10. Panguluri, S. K., & Kalivarapu, V. (2017). "A smart irrigation system using Arduino and IoT." *International Journal of Innovations in Engineering and Technology (IJIET)*, 8(4), 36–40.
11. Gorantiwar, S. D., Smout, I. K., & Patel, A. J. (2012). "Decision support system for planning and design of micro-irrigation systems." *Biosystems Engineering*, 113(3), 319–331.
12. Jones, H. G. (2004). *Irrigation Scheduling: Advantages and Pitfalls of Plant-Based Methods*. FAO Water Reports.
13. Martínez-Romero, A., et al. (2020). "Water-saving irrigation systems based on soil moisture monitoring." *Sensors*, 20(20), 5965.
14. Morais, R., Fernandes, M. A., & Mendes, J. (2014). "A ZigBee multi-powered wireless acquisition device for remote sensing applications in precision viticulture." *Computers and Electronics in Agriculture*, 62(2), 94–106.
15. Tripathy, B., & Ahirwar, S. (2020). "Role of machine learning and artificial intelligence in precision agriculture." *Handbook of IoT and Big Data Technologies*, 1057–1078.
16. Islam, M. R., Islam, M. F., & Hoque, M. A. (2020). "Solar powered smart irrigation system using Arduino." *Journal of Computing and Security*, 5(2), 45–55.